

Cryptocurrencies

At the intersection of the worlds of finance, the internet, and cryptography are cryptocurrencies. These are digital currencies that have been causing a stir in since 2009.

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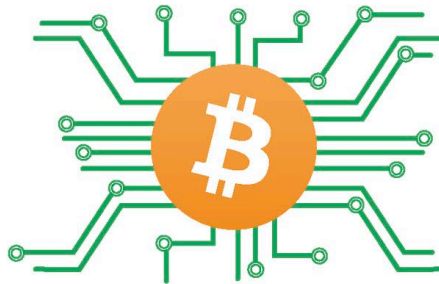
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What are they?

A cryptocurrency is a fully digital form of currency. The first, bitcoin, was created in 2009, but there are currently more than 1,000 different ones. Cryptocurrencies are decentralized: they are not created or overseen by one government or bank. They make it possible to send money to others or pay for things from anywhere in the world, without needing to convert money into local currency. Cryptocurrency transactions are made extremely secure with cryptography.

▷ Bitcoin

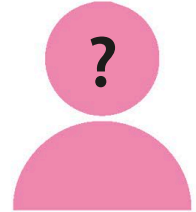
Bitcoin is the first and most valuable cryptocurrency. A satoshi – named after bitcoin's creator Satoshi Nakamoto – is the smallest possible amount of bitcoin. There are 100,000,000 satoshis in one bitcoin.



BIOGRAPHY

Satoshi Nakamoto

The person who created bitcoin in 2009 gave his name as Satoshi Nakamoto, and claimed to have been born in Japan in 1975. Aside from being active in the development of bitcoin until late 2010, not much else is known about Nakamoto. We don't know if he or she is a real person, or a pseudonym. We don't know if he or she is even Japanese. What we do know is that the person who set up the system has roughly one million bitcoins.



How bitcoin works

Each cryptocurrency works in its own way, but many follow the lead of bitcoin. For a bitcoin transaction to work, both parties need to have a wallet (where currency can be stored) and an address (a string of letters and numbers that are used to send currency to and from).



△ Wallets and addresses

Bitcoin is stored in a user's wallet. When making a transaction, users are encouraged to create a new address to maintain their anonymity.



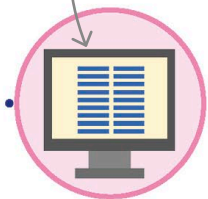
△ Bitcoin clients

Users can then make a payment by using their computer or a mobile device. All they need is a bitcoin program, called a client.



△ Miners

The payment is then verified by miners. Miners are the people who maintain bitcoin. They check that someone spending bitcoins has enough for the transaction.



△ The blockchain

Once a transaction is made, it is added to the blockchain, the list of all confirmed bitcoin transactions.

Miners use complex cryptography to ensure the transaction is possible, and receive a small amount of bitcoin in return.

Everybody can see the list of bitcoin transactions on the blockchain.